

Carver Mead, an American scientist and engineer.

Fast forward to today

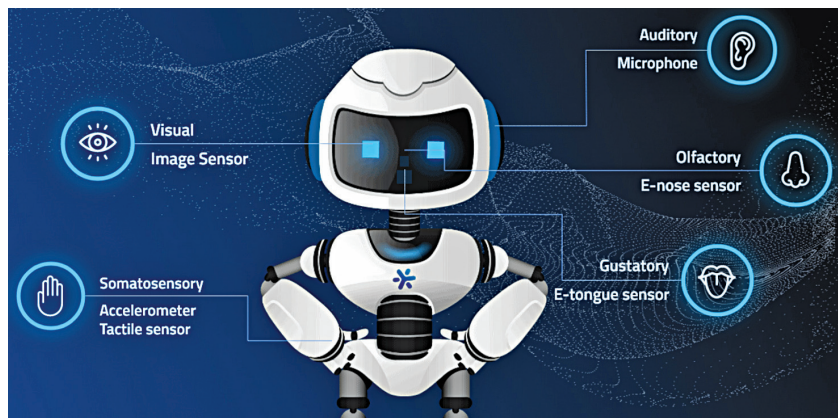
Nowadays, almost all companies have dedicated teams working on neuromorphic computing. Groundbreaking research is being done in multiple research organisations and universities. It is safe to say that neuromorphic computing is gaining momentum and will continue to do so as various advancements are being made.

What's interesting to note is that although this is a specialised field with prerequisites from various topics, including solid-state physics, VLSI, neural networks, and computational neurobiology, undergraduate engineering students are extremely curious about this field.

At IIT Kanpur, Dr Shubham Sahay, Assistant Professor at the Department of Electrical Engineering, introduced a course on neuromorphic computing last year. Despite being a post-graduate level course, he saw great participation from undergrads as well. "Throughout the course, they were very interactive. The huge B.Tech participation in my course bears testimony to the fact that undergrads are really interested in this topic. I believe that this (neuromorphic computing) could be introduced as one of the core courses in the UG curriculum in the future," he says.

Getting it commercial

Until recently, neuromorphic computing was a widely used term only in research and not in the commercial arena. However, as of January 18, 2022, BrainChip, a leading provider of ultra-low-power high-performance AI technology, commercialised its AKD1000 AIOT chip. Developers, system integrators, and engineers can now buy AKD1000-powered Mini PCIe boards and leverage them in their applications, especially those requiring on-edge computing, low power consumption, and high-performance AI.



The five sensor modalities (Image Source: BrainChip Inc.)

"It's meant as our entry-level product. We want to proliferate this into as many hands as we can and get people designing in the Akida environment," says Rob Telson, Vice President of WorldWide Sales at BrainChip. Anil Mankar, Co-founder and Chief Development Officer of BrainChip, explains, "We are enabling system integrators to easily use neuromorphic AI in their applications. In India, if some system integrators want to manufacture the board locally, they can take the bill of materials from us (BrainChip) and manufacture it locally."

What's fascinating about Akida is that it enables sensor nodes to compute without depending on the cloud. Further, BrainChip's AI technology not only performs audio and video based learning but even focuses on other sensor modalities like taste, vibration, and smell. You can use it to make a sensor that performs wine tasting! Here is a link to their wine tasting demonstration: <https://youtu.be/RwWTietRioM>

Another major event that occurred this year was when Mercedes implemented BrainChip's Akida technology in its Vision EQXX electric vehicle. This is definitely a big deal since the Akida technology is tried and tested for a smart automotive experience. All features that the Akida provides, including facial recognition,

keyword spotting, etc, consume extremely low power.

"This is where we get excited. You'll see a lot of these functionalities in vehicles—recognition of voices, faces, and individuals in the vehicle. This allows the vehicles to have customisation and device personalisation according to the drivers or the passengers as well," says Telson. These really are exciting times.

Neuromorphic hardware, neural networks, and AI

The process in which neurons work is eerily similar to an electric process. Neurons communicate with each other via synapses. Whenever they receive input, they produce electrical signals called spikes (also called action potentials), and the event is called neuron spiking. When this happens, chemicals called neurotransmitters are released into hundreds of synapses and activate the respective neurons. That's the reason why this process is super-fast.

Artificial neural networks mimic the logic of the human brain, but on a regular computer. The thing is, regular computers work on the Von Neumann architecture, which is extremely different from the architecture of our brain and is very power-hungry. We may not be able to deploy CMOS logic on the Von Neumann architecture for long. We will